

ORB (Oriented FAST and Rotated BRIEF) for Stop Signs

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Background & Project Proposal

Background

- Road signs can be missed in urban environments due to clutter
- In urban environments there are many different road signs which can cause false positive detections
- It is challenging to perform object detection when the environment is crowded
- Applications
 - Self driving vehicles
 - Vehicle navigation
 - Smart glasses



Problem Statement

“We will use ORB (Oriented Fast and Rotated Brief) to detect distinctive features and attributes of stop signs and then use the iterative RANSAC (Random Sample Consensus) algorithm to identify matched keypoints between the referenced image and a provided images and videos. This will allow us to detect and then track the stop signs through the frames of the video.”

Reference Paper: E. Rublee, V. Rabaud, K. Konolige, and G. Bradski,
“ORB: An efficient alternative to SIFT or SURF,” in Proc. IEEE Int. Conf.
Comput. Vis. (ICCV), Barcelona, Spain, Nov. 2011

Relation to Course

- Perception unit
 - “Building a model of the world from sensor data”
- Realistic robot perception problem
 - Understanding of environment
- ORB
 - Applications in SLAM
- RANSAC
 - Outlier rejection
 - Geometric Transformation Matrix
- Interest in computer vision!

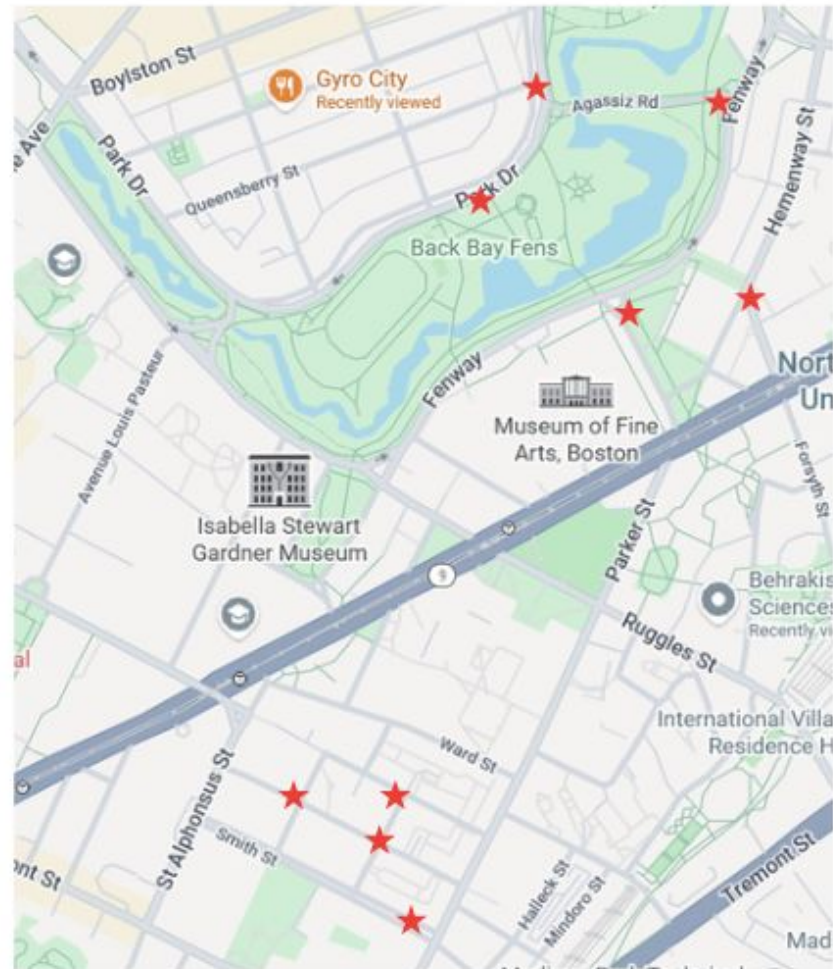
3 + 1 Pillars of (Robotics) Wisdom



Lecture 1 Image!

Data Acquisition Map

- We acquired pictures & videos of signs around Boston at these locations on the map
- Tried to find signs in different environments and backgrounds to accurately test the algorithm



Proposed Experiment

- Pictures 45 Images Total at 9 stop signs
 - 8 Close up: 3 feet away level with the sign
 - 9 Angled: close up but at a lower angle
 - 9 Far away: 20 feet away level with sign
 - 10 signs that aren't stop signs 3 feet away level with the sign
 - 10 images on streets without any signs
- Videos (6 Videos Total)
 - 4 driving at 10mph past a stop sign
 - 1 with sign that isn't stop sign
 - 1 with no sign at all



Proposed Solution

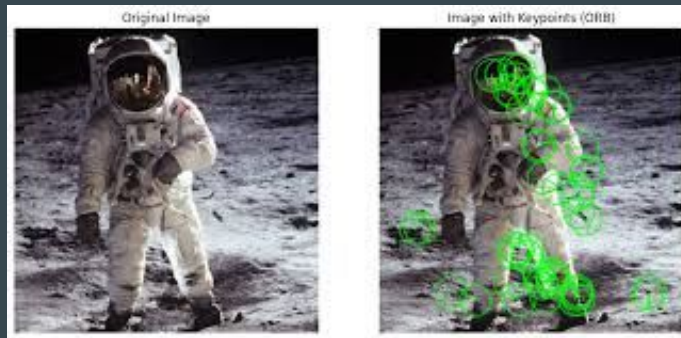
ORB Overview: Oriented FAST and Rotated BRIEF

First part - Oriented FAST (Features from Accelerated Segment Test)

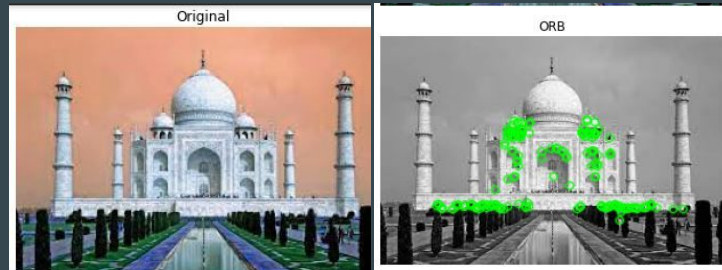
- Detects corners (key points) by checking a circle of pixels around a point
- Gives each keypoint a rotation angle by computing the intensity centroid - essentially a center of mass of brightness

Second Stage - Rotated Brief

- Uses that rotation angle to align the descriptor (fingerprint) for that pixel area to align with the keypoint's orientation
- Captures pattern of light/dark structure around a corner



ORB Feature Detection



ORB Improvements from FAST & BRIEF

oFAST Expands on traditional FAST to account for rotation

- Uses Image Moment: weighted sum of pixel intensities
- Computes Centroid: center of mass of intensities, average of brightness around the keypoint
- Computes Orientation Angle: finds the angle of centroid orientation

rBRIEF Expands on traditional BRIEF to with rotation Matrix for keypoint matching

$$m_{pq} = \sum_x \sum_y x^p y^q I(x, y)$$

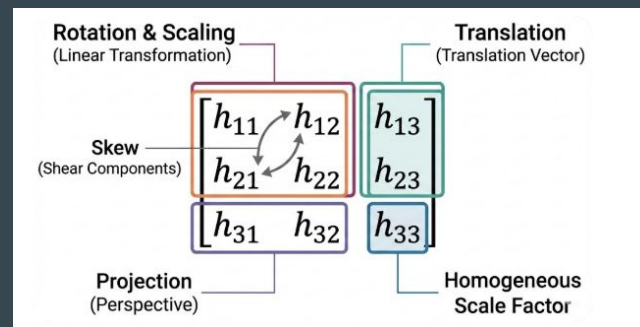
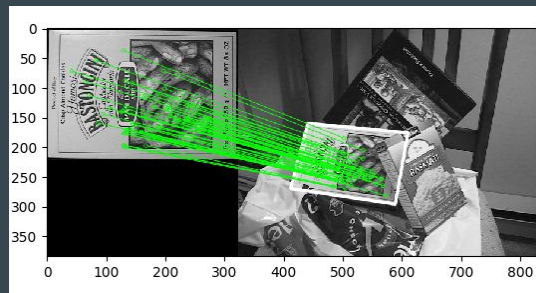
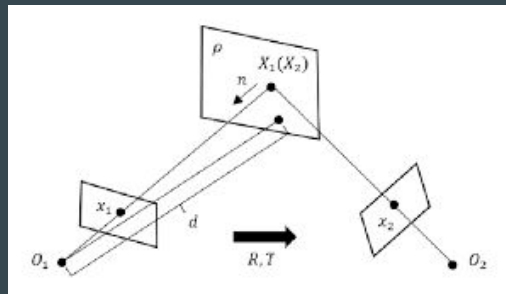
$$C = \left(\frac{m_{10}}{m_{00}}, \frac{m_{01}}{m_{00}} \right)$$

$$\theta = \text{atan2}(m_{01}, m_{10})$$

$$S_\theta = R_\theta S$$

RANSAC Overview: RANdom SAmple Consensus

- Takes all matched keypoint pairs from reference and test picture and compares their relation to each other
- Generates a homography matrix that defines the “rule” for - translation, rotation, scaling, warping
- All key points should adhere to this “rule”, if not, they’re not an inlier match



Our Algorithm - Detect Stop Sign Function

1. Resizes and formats image/video for processing
 - a. Resizes & Adds Gaussian Blur
2. Uses ORB for initial feature detection in reference and tests images
3. Uses Brute Force matcher with Hamming distance to match features between reference and test image
4. Apply RANSAC to find relationship/translation between the two images
5. Use Homography matrix to warp the first image to align with second image - determine inliers
6. Identifies or does not identify stop sign based on inlier threshold set

Picture Results

Picture Looping: Quantifying Results

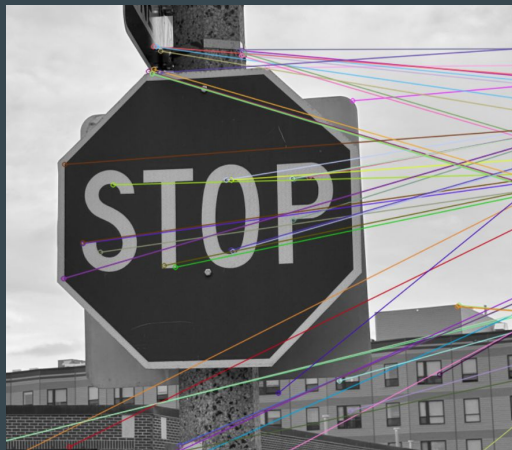
We have 5 folders each with 8-10 pictures within these categories:

- Stop Sign - Close up, 4-5 feet away head on
- Stop Sign - Far, 20-25 feet away
- Stop Angle - 4-5 feet away, at an angle
- No Signs
- Other Road Signs

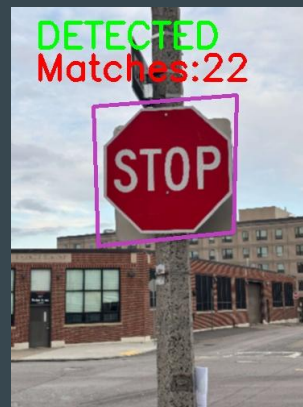
We loop through each folder and run the algorithm on each image while keeping track of accuracy per image & saving output results



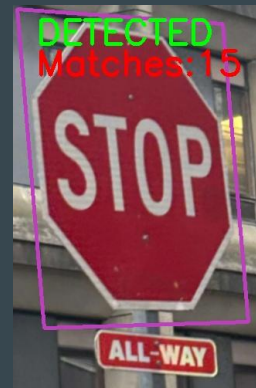
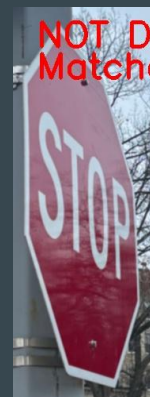
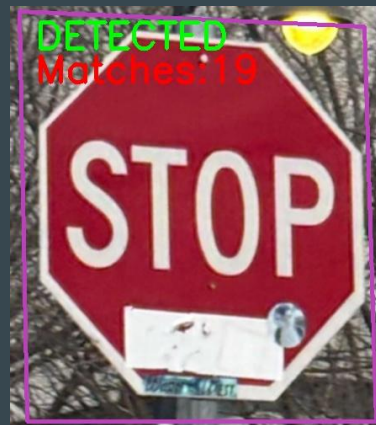
Feature Detection and Matching Examples



Stop Close 85.71% Accuracy



Stop Angled 55.56% Accuracy



Stop Far 0% Accuracy



No Stop 100% Accuracy



Other Signs 100% Accuracy



Results Analysis

Paper Findings

- Strong rotation invariance
- Fast and stable keypoints
- Poor scale invariance

Our Results

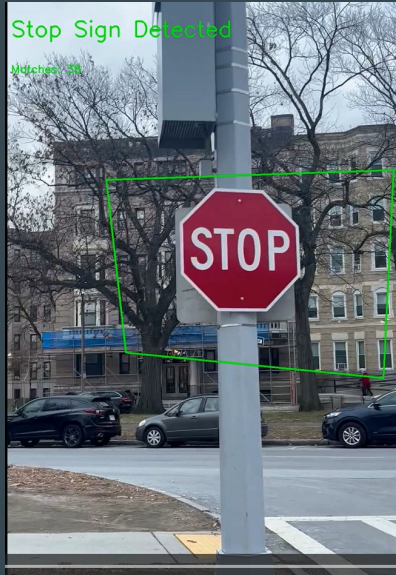
- Angled detection works (55%)
- Good close-up detection (85%)
- Total failure at 20 ft (0%)

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FINAL ACCURACY SUMMARY
stop-close: 85.71%
stop-angle: 55.56%
stop-far: 0.00%
no-stop: 100.00%
other-sign: 100.00%
```

Video Results

Video Results

- We had to reduce the video resolution for simpler computation
- Added checks for aspect ratio and relative feature size to account for movement throughout the video



Video Results

- Processes frame by frame
- Relatively successful but only in the intermediate not too close - not too far range
 - and only in cases without glare
- Was able to correctly predict even with angle changes through camera movement
- Could not move too fast, the motion blur affected the algorithm's ability to distinguish features

Do Not Enter Sign



Stop Sign 1



Stop Sign 2



Conclusion

Limitations & Implementation Challenges

- Preprocessing
 - Heavy reliance on picture quality
- Reliance on setting thresholds
 - We had to optimize values and fine tuned inlier count threshold and image quality for best results
- Randomness
 - Results varied from run to run because RANSAC samples random variables
 - Can cause varied results with non optimal or unstable homography estimates

Conclusion & Future Work

Conclusion

- ORB & RANSAC provide a fast approach for stop sign detection
- Performance is very dependent on image quality and scaling

Future Work

- More advanced detection algorithms or image processing techniques would be necessary to improve accuracy
- New image processing techniques to improve accuracy in low visibility
- Expand dataset to include different images
 - Time of day: night time visibility
 - Weather variety: rain, snow, fog, etc

References

[1] E. Rublee, V. Rabaud, K. Konolige, and G. Bradski,

“ORB: An efficient alternative to SIFT or SURF,” in Proc. IEEE Int. Conf. Comput. Vis. (ICCV), Barcelona, Spain, Nov. 2011, pp. 2564–2571.

[2] GeeksforGeeks,

“Step-by-step guide to using RANSAC in OpenCV using Python,”

<https://www.geeksforgeeks.org/python/step-by-step-guide-to-using-ransac-in-opencv-using-python/> (Accessed: Dec. 1, 2025).